

False positive rate

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In statistics, when performing multiple comparisons, the term **false positive ratio**, also known as the **false alarm ratio**, usually refers to the probability of falsely rejecting the null hypothesis for a particular test.

The false positive **rate** (or "false alarm rate") usually refers to the **expectancy** of the false positive **ratio**.

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Quick Definition

The false positive rate is $\frac{FP}{FP + TN}$.

Where FP is number of false positives, and TN is number of true negatives.

Definition

Suppose we have m null hypotheses, denoted by: $H_1, H_2, ..., H_m$.

Using a statistical test, each hypothesis is declared significant/non-significant.

Summing the test results over H_i will give us the following table and related random variables:

	Null hypothesis is True	Alternative hypothesis is True	Total
Declared significant	V	S	R
Declared non-significant	U	T	$m - R$
Total	m_0	$m - m_0$	m

- m_0 is the number of true null hypotheses, an unknown parameter
- $m - m_0$ is the number of true alternative hypotheses
- V is the number of false positives (Type I error)
- S is the number of true positives
- T is the number of false negatives (Type II error)
- U is the number of true negatives

- R is the number of rejected null hypotheses. It is an observable random variable, while S, T, U , and V are unobservable random variables.

The level of significance that is used to test each hypothesis is set based on the form of inference (simultaneous inference vs. selective inference) and its supporting criteria (for example FWER or FDR), that were pre-determined by the researcher.

When performing multiple comparisons in a statistical framework such as above, the **false positive ratio** (also known as the **false alarm ratio**, as opposed to false positive **rate** / false alarm **rate**) usually refers to the probability of falsely rejecting the null hypothesis for a particular test. Using the terminology suggested here, it is simply V/m_0 .

Since V is a random variable and m_0 is a constant ($V \leq m_0$), the false positive **ratio** is also a random variable, ranging between 0-1.

The **false positive rate** (or "false alarm rate") usually refers to the **expectancy of the false positive ratio**, expressed by $E(V/m_0)$.

It is worth noticing that the two definitions ("false positive ratio" / "false positive rate") are somewhat interchangeable. For example, in the referenced article^[1] V/m_0 serves as the false positive "rate" rather than as its "ratio".

The difference between "false positive rate", "type I error rate" and other close terms

While the false positive rate is mathematically equal to the type I error rate, it is viewed as a separate term for the following reasons:

- The type I error rate is often associated with the **a-prior** setting of the significance level by the researcher: the significance level represents an acceptable error rate **considering that all null hypotheses are true** (the "global null" hypothesis). the choice of a significance level may thus be somewhat arbitrary (i.e. setting 10% (0.1), 5% (0.05), 1% (0.01) etc.)

As opposed to that, the false positive rate is associated with a **post-prior** result, which is the expected number of false positives divided by the total number of hypotheses under the **real** combination of true and non-true null hypotheses (disregarding the "global null" hypothesis). Since the false positive rate is a parameter that is not controlled by the researcher, it cannot be identified with the significance level.

- Moreover, false positive rate is a term usually used regarding a medical test or diagnostic device (i.e. "the false positive rate of a certain diagnostic device is 1%"), while type I error is a term associated with statistical tests, where the meaning of the word "positive" is not as clear (i.e. "the type I error of a test is 1%").

The false positive rate should also not be confused with the familywise error rate, which is defined as $\text{FWER} = \Pr(V \geq 1)$. As the number of tests grows, the familywise error rate usually converges to 1 while the false positive rate remains fixed.

Lastly, it is important to note the profound difference between the false positive rate and the false discovery rate: while the first is defined as $E(V/m_0)$, the second is defined as $E(V/R)$.

References

1. Burke, Donald; Brundage, John; Redfield, Robert (1988). "Measurement of the False Positive Rate in a Screening Program for Human Immunodeficiency Virus Infections". *The New England Journal of Medicine* **319**: 961–964. doi:10.1056/NEJM198810133191501. PMID 3419477.

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